

QUALITY ASSURANCE AUDIT REPORT.

REVIEWING COVERAGE, NOT JUST THE PASS RATE

The Day 4 QA suite reports 6/6 tests passed. This poster checks what those 6 tests actually cover — and what they don't.

REPORTED RESULT

6 / 6 PASSED

100% pass rate on the checks that were implemented. All 6 are genuine, verifiable passes.

2,038 0
ROWS VALIDATED MISSING / DUPES

WHAT WAS TESTED

- ✓ **Dataset Availability** — loads as DataFrame
- ✓ **Dataset Not Empty** — 2,038 rows
- ✓ **Duplicate Check** — 0 duplicates
- ✓ **Missing Values Check** — 0 nulls
- ✓ **Cluster Column** — present
- ✓ **KMeans Model File** — found & loads

FILE & MODEL FACTS

Model expects **40 features**, all confirmed present in the dataset. day3_results.pkl contains all **17 expected keys**.

Income sample value: 0.299651 — still standardized, not real dollars.

THE HEADLINE OVERSTATES CONFIDENCE

All 6 checks are **file-exists, shape, and null checks** — not functional or semantic tests. None of them would catch the production-blocking bugs already documented elsewhere in this project: the empty marketing strategy matrix, the z-score dollar display bug, or the non-functional prediction page.

WHAT A RIGOROUS PASS WOULD ALSO CHECK

Check	Why It's Missing
Prediction consistency	No test re-runs model.predict() against the stored Cluster column — no proof labels came from this model.
Streamlit output testing	Named explicitly in the Day 4 header, but zero cells touch app.py — the one promised category never tested.
Semantic value checks	17/17 keys "exist" in day3_results.pkl, but values are never inspected — would still PASS on an empty table.
Unit / scale check	Income is still a z-score; nothing flags this, though it's the exact cause of the Module 10 dashboard bug.
Segment-name consistency	No check for which of 3 historical cluster names is present in this file.

MINOR NOTEBOOK ISSUES

- Model-loading cell duplicated verbatim (copy-paste artifact)
- An out-of-order check references results_test_df before it's created — always fails harmlessly
- day3_results.pkl's own folder path mixes 3 different week/module labels

HONEST HEADLINE

Not "100% pass, pipeline is production-ready" — more accurately: **"100% pass on 6 of roughly 15 recommended checks."** The gap is coverage, not correctness of what was tested.

PRIORITY FIX LIST

- 1 **Prediction-consistency test** — assert model.predict() matches stored Cluster labels
- 2 **Add the missing Streamlit test** — smoke-test app.py, assert inverse-transform before display
- 3 **Value-level checks** on all 17 day3_results.pkl keys, not just key existence
- 4 **Explicit scale/units assertion** on Income and spending columns
- 5 **Report coverage alongside pass rate** in every QA summary

WHY THIS MATTERS

A QA suite's job is to catch regressions *before* they reach users. This one would have let the Module 10 dashboard bug and the Module 8 empty-matrix bug both ship with a green checkmark.

